## Exercise J-M

1. The equations of the sides AB and AC of a triangle ABC are $(\lambda + 1)x + \lambda y = 4$ and $\lambda x + (1 - \lambda)\, y + \lambda = 0$ respectively. Its vertex A is on the y-axis and its orthocentre is (1, 2). The length of the tangent from the point C to the part of the parabola $y^2 = 6x$ in the first quadrant is : **[JEE(Main)Jan.2023]**

एक त्रिभुज ABC की भुजाओं AB और AC के समीकरण क्रमशः $(\lambda + 1)x + \lambda y = 4$ और $\lambda x + (1 - \lambda)\, y + \lambda = 0$ है। इसका शीर्ष A, y-अक्ष पर है और इसका लंबकेन्द्र (1, 2) है। बिंदु C से परवलय $y^2 = 6x$ के प्रथम चतुर्थांश के भाग पर खींची गई स्पर्श रेखा की लंबाई है : **[JEE(Main)Jan.2023]**

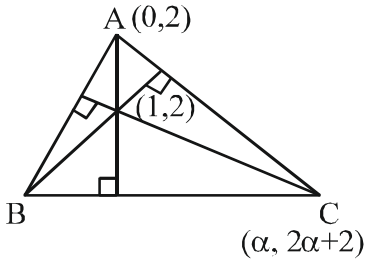
(A) $2\sqrt{2}$ (B) 2 (C) 4 (D) $\sqrt{6}$

**Ans.** A

**Sol.** $AB : (\lambda+1)x + \lambda y = 4$

$AC : \lambda x + (1-\lambda)y + \lambda = 0$

Vertex A is on y-axis




$\Rightarrow x = 0$

$y = \frac{4}{\lambda}, y = \frac{\lambda}{\lambda - 1}$

$\Rightarrow \frac{4}{\lambda} = \frac{\lambda}{\lambda - 1}$

$\Rightarrow \lambda = 2$

$AB : 3x + 2y = 4$

$AC : 2x - y + 2 = 0$

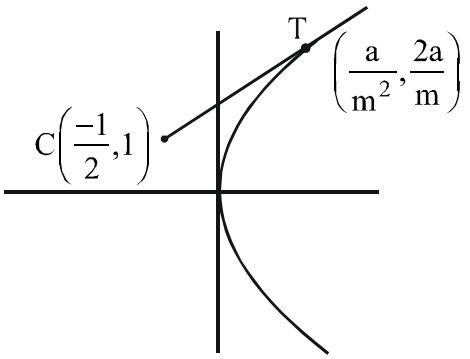
$\Rightarrow A(0, 2)$

Let $C(\alpha, 2\alpha + 2)$

Now (Slope of Altitude through C) $\left(-\frac{3}{2}\right) = -1$

$\left(\frac{2\alpha}{\alpha - 1}\right)\left(-\frac{3}{2}\right) = -1 \Rightarrow \alpha = -\frac{1}{2}$

So $C\left(-\frac{1}{2}, 1\right)$




Let Equation of tangent be $y = mx + \frac{3}{2m}$

$m^2 + 2m - 3 = 0$

$\Rightarrow m = 1, -3$

So tangent which touches in first quadrant at T is

$T \equiv \left(\frac{a}{m^2},\frac{2a}{m}\right)$

$\equiv \left(\frac{3}{2},3\right)$

$\Rightarrow CT = \sqrt{4+4} = 2\sqrt{2}$

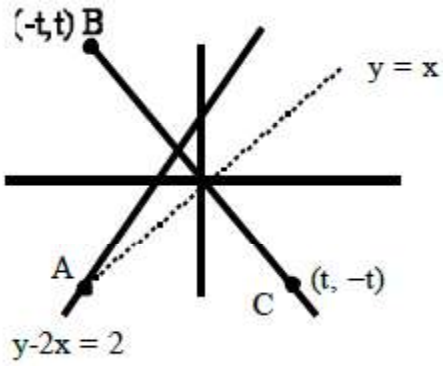
2. Let B and C be the two points on the line $y + x = 0$ such that B and C are symmetric with respect to the origin. Suppose A is a point on $y - 2x = 2$ such that $\Delta ABC$ is an equilateral triangle. Then, the area of the $\Delta ABC$ is : **[JEE(Main)Jan.2023]**

माना रेखा $y + x = 0$ पर दो बिंदु B व C मूल बिंदु के सममित है। माना $y - 2x = 2$ पर एक बिंदु A इस प्रकार है कि $\Delta ABC$ एक समबाहु त्रिभुज है। तब $\Delta ABC$ का क्षेत्रफल है : **[JEE(Main)Jan.2023]**

(A) $3\sqrt{3}$ (B) $2\sqrt{3}$ (C) $\frac{8}{\sqrt{3}}$ (D) $\frac{10}{\sqrt{3}}$

**Ans.** C

**Sol.**



At A $x = y$

$Y - 2x = 2$

$(-2, -2)$

Height from line $x + y = 0$

$$h = \frac{4}{\sqrt{2}}$$

$$\text{Area of } \Delta = \frac{\sqrt{3}}{4}\frac{h^2}{\sin^2 60} = \frac{8}{\sqrt{3}}$$

3. A light ray emits from the origin making an angle $30^o$ with the positive x-axis. After getting reflected by the line $x + y = 1$, if this ray intersects x-axis at Q, then the abscissa of Q is : **[JEE(Main)Jan.2023]**

X-अक्ष की धनात्मक दिशा से $30^o$ का कोण बनाती हुई मूल बिंदु से एक प्रकाश किरण उत्सर्जित होती है। रेखा $x + y = 1$ से परावर्तित होने के बाद, यदि यह किरण x-अक्ष को Q पर मिलती है, तब Q का भुज है : **[JEE(Main)Jan.2023]**

(A) $\frac{2}{3-\sqrt{3}}$ (B) $\frac{\sqrt{3}}{2\left(\sqrt{3}+1\right)}$ (C) $\frac{2}{\left(\sqrt{3}-1\right)}$ (D) $\frac{2}{3+\sqrt{3}}$

**Ans.** D

**Sol.** Image of O (0, 0) in line $x + y - 1 = 0$ lies on reflected ray.

$$\frac{x-0}{1} = \frac{y-0}{1} = \frac{-2(0+0-1)}{2} \Rightarrow B\ (1, 1).$$

Also, upon solving $y = \frac{x}{\sqrt{3}}$ and $x + y - 1 = 0$ we get $P \equiv \left(\frac{3-\sqrt{3}}{2}, \frac{\sqrt{3}-1}{2}\right)$

equation of reflected ray is same as line passing through BP.

$$\text{slope} = \frac{\frac{\sqrt{3}}{2} - \frac{1}{2} - 1}{\frac{3}{2} - \frac{\sqrt{3}}{2} - 1} = \frac{\sqrt{3}-3}{1-\sqrt{3}} = \sqrt{3}$$

Equation of line BP is

$y - 1 = \sqrt{3}(x - 1)$

Put $y = 0 \Rightarrow -\frac{1}{\sqrt{3}} = x - 1$

Required point $\left(1 - \frac{1}{\sqrt{3}}, 0\right)$

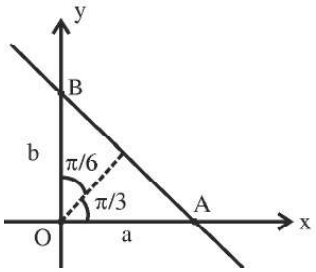
4. A straight line cuts off the intercepts OA = a and OB = b on the positive directions of *x*-axis and *y*-axis respectively. If the perpendicular from origin O to this line makes an angle of $\frac{\pi}{6}$ with positive direction of *y*-axis and the area of $\Delta OAB$ is $\frac{98}{3}\sqrt{3}$, then $a^2 - b^2$ is equal to : **[JEE(Main)Jan.2023]**

एक सरल रेखा x-अक्ष तथा y-अक्ष की घनात्मक दिशाओं पर क्रमशः OA = a तथा OB = b अंतःखंड काटती है। यदि मूलबिंदु से इस रेखा पर अभिलंब y-अक्ष की घनात्मक दिशा से $\frac{\pi}{6}$ का कोण बनाता है तथा $\Delta OAB$ का क्षेत्रफल $\frac{98}{3}\sqrt{3}$ है, तो $a^2 - b^2$ बराबर है : **[JEE(Main)Jan.2023]**

(A) 98 (B) $\frac{196}{3}$ (C) 196 (D) $\frac{392}{3}$

**Ans.** D

**Sol.**




Equation of straight line : $\frac{x}{a} + \frac{y}{b} = 1$

Or $x\cos\frac{\pi}{3} + y\sin\frac{\pi}{3} = p$

$\frac{x}{2} + \frac{y\sqrt{3}}{2} = p$

$\frac{x}{3p} + \frac{y}{2p} = 1$

Comparing both : $a = 2p,\ b = \frac{2p}{\sqrt{3}}$

Now area of $\Delta OAB = \frac{1}{2}\cdot ab = \frac{98}{3}\cdot\sqrt{3}$

$\frac{1}{2}\cdot 2p\cdot\frac{2p}{\sqrt{3}} = \frac{98}{3}\cdot\sqrt{3}$

$p^2 = 49$

$$a^2 - b^2 = 4p^2 - \frac{4p^2}{3} = \frac{2}{3}4p^2$$

$$= \frac{8}{3} \cdot 49 = \frac{392}{3}$$

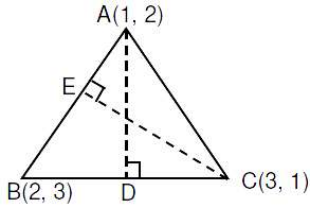
5. If the orthocenter of the triangle, whose vertices are (1, 2), (2, 3) and (3, 1) is $(\alpha, \beta)$, then the quadratic equation whose roots are $\alpha + 4\beta$ and $4\alpha + \beta$, is : **[JEE(Main)Jan.2023]**

एक त्रिभुज के शीर्ष (1, 2), (2, 3) और (3, 1) है। यदि इसका लंबकेन्द्र $(\alpha, \beta)$ है, तब $\alpha + 4\beta$ और $4\alpha + \beta$ किस समीकरण के मूल है : **[JEE(Main)Jan.2023]**

(A) $x^2 - 19x + 90 = 0$ (B) $x^2 - 20x + 99 = 0$ (C) $x^2 - 18x + 80 = 0$ (D) $x^2 - 22x + 120 = 0$

**Ans.** B

**Sol.**




$$\text{Slope (AB)} = m_{AB} = \frac{2-3}{1-2} = 1$$

$\therefore$ Slope (CE) $= -1$

$\therefore$ Equation of CE is : $y - 1 = -1(x - 3)$

$\Rightarrow y - 1 = -x + 3$

$\Rightarrow x + y = 4$ ...(i)

$$\text{Similarly } m_{BC} = \frac{2}{-1} = -2$$

$$\text{Equation of AD is : } y - 2 = \frac{1}{2}(x - 1)$$

$2y - 4 = x - 1$

$\therefore x - 2y + 3 = 0$ ...(ii)

Solving equation (i) & (ii)

$$\text{We have } x = \frac{5}{3},\ y = \frac{7}{3}$$

$$\therefore \alpha = \frac{5}{3}, \beta = \frac{7}{3}$$

$$\therefore \alpha + 4\beta = \frac{5}{3} + \frac{28}{3} = \frac{33}{3} = 11$$

$4\alpha+\beta=4\times\frac{5}{3}+\frac{7}{3}=\frac{27}{3}=9$

$\therefore$ Quadratic Equation is

$x^2-(\text{sum of roots})x+(\text{products of roots})=0$

$x^2-20x+99=0$

6. The combined equation of the two lines $ax+by+c=0$ and $a'x+b'y+c'=0$ can be written as $(ax+by+c)\left(a'x+b'y+c'\right)=0$. The equation of the angle bisectors of the lines represented by the equation $2x^2+xy-3y^2=0$ is : **[JEE(Main)Jan.2023]**

दो रेखाओं $ax+by+c=0$ तथा $a'x+b'y+c'=0$ का संयुक्त समीकरण $(ax+by+c)\left(a'x+b'y+c'\right)=0$ लिखा जा सकता है। समीकरण $2x^2+xy-3y^2=0$ से प्राप्त दो रेखाओं के कोण समद्विभाजकों का समीकरण है। **[JEE(Main)Jan.2023]**

(A) $x^2-y^2-10xy=0$ (B) $3x^2+5xy+2y^2=0$

(C) $x^2-y^2+10xy=0$ (D) $3x^2+xy-2y^2=0$

**Ans.** A

**Sol.** Equation of angle bisectors is $\frac{x^2-y^2}{2+3}=\frac{xy}{\frac{1}{2}}$

$\Rightarrow\frac{x^2-y^2}{5}=2xy$

$\Rightarrow x^2-y^2=10xy$

$\Rightarrow x^2-y^2-10xy=0$

7. The equations of the sides AB, BC and CA of a triangle ABC are : $2x+y=0$, $x+py=21a$, $(a\neq0)$ and $x-y=3$ respectively. Let P(2, a) be the centroid of $\Delta ABC$. Then $(BC)^2$ is equal to _____. **[JEE(Main)Jan.2023]**

एक त्रिभुज ABC की भुजाओं AB, BC तथा CA के समीकरण क्रमशः $2x+y=0$, $x+py=21a$, $(a\neq0)$ तथा $x-y=3$ है। माना $\Delta ABC$ का केन्द्रक P(2, a) है। तब $(BC)^2$ बराबर है। **[JEE(Main)Jan.2023]**

**Ans.** 122

**Sol.** Now $\frac{1+\alpha+\beta}{3}=2$

$\alpha+\beta=5\ldots(1)$

and $\frac{-2-2\beta+\alpha-3}{3}=a\Rightarrow\alpha-2\beta=3a+5$ _____(2)

from (1) and (2) $\alpha = a + 5, \beta = -a$

B( – a, 2 a), C(a + 5, a + 2)

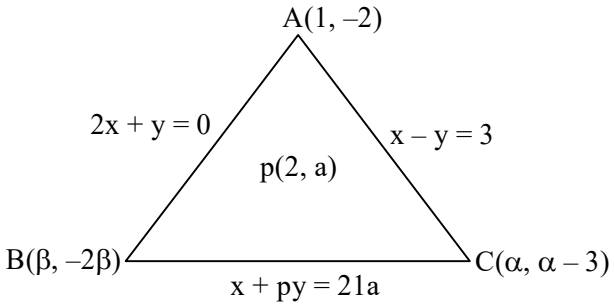
B and C lies on x + py = 21a

–a + p(2a) = 21a $\Rightarrow$ p = 11

and a + 5 + 11(a + 2) = 21a

$\Rightarrow$ a = 3

B(–3, 6), C(8, 5), $BC^2 = 122$

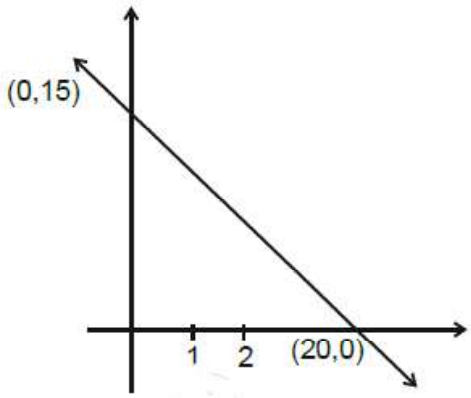



8. A triangle is formed by X-axis, Y-axis and the line $3x + 4y = 60$. Then the number of points P(a,b) which lie strictly inside the triangle, where a is an integer and b is a multiple of a, is _____. **[JEE(Main)Jan.2023]**

X-अक्ष, Y-अक्ष तथा रेखा $3x + 4y = 60$ एक त्रिभुज बनाते है। तो ऐसे बिंदुओं P(a,b) जहाँ a एक पूर्णांक है तथा b, a का एक गुणज है, जो त्रिभुज के अंदर है, की संख्या है। **[JEE(Main)Jan.2023]**

**Ans.** 31

**Sol.** If $x = 1, y = \frac{57}{4} = 14.25$




(1, 1) (1, 2) – (1, 14) $\Rightarrow$ 14 pts.

If $x = 2,\ y = \frac{27}{2} = 13.5$

(2, 2) (2, 4) ¡K. (2, 12) $\Rightarrow$ 6 pts.

If $x = 3,\ y = \frac{51}{4} = 12.75$

(3, 3) (3, 6) – (3, 12) $\Rightarrow$ 4 pts.

If x = 4, y = 12

(4, 4) (4, 8) $\Rightarrow$ pts.

If $x = 5.\ y = \frac{45}{4} = 11.25$

(5, 5), (5, 10) $\Rightarrow$ 2 pts.

If $x = 6,\ y = \frac{21}{2} = 10.5$

(6, 6) $\Rightarrow$ 1 pt.

If $x = 7,\ y = \frac{39}{4} = 9.75$

(7, 7) $\Rightarrow$ 1 pt.

If $x = 8,\ y = 9$

(8, 8) $\Rightarrow$ 1 pt.

If $x = 9\ \ y = \frac{33}{4} = 8.25$ $\Rightarrow$ no pt.

Total = 31 pts.

9. The straight lines $l_1$ and $l_2$ pass through the origin and trisect the line segment of the line $L : 9x + 5y = 45$ between the axes. If $m_1$ and $m_2$ are the slopes of the lines $l_1$ and $l_2$ then the point of intersection of the line $y = (m_1 + m_2)x$ with L lies on : **[JEE(Main)April,2023]**

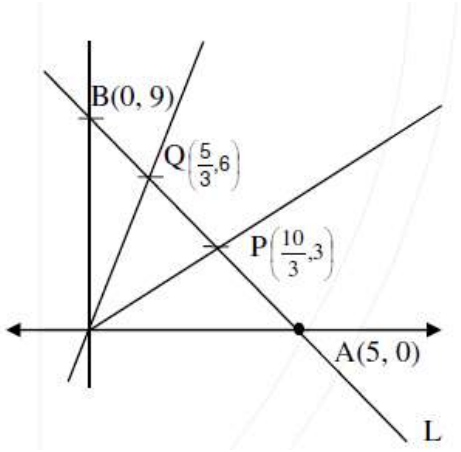
सरल रेखाएँ $l_1$ व $l_2$ मूल बिंदु से होकर जाती है और रेखा $L : 9x + 5y = 45$ के अक्षों के बीच रेखाखंड को तीन बराबर भागों में बाँटती है। यदि रेखाओं $l_1$ व $l_2$ की प्रवणताएँ $m_1$ व $m_2$ हैं, तब रेखाओं $y = (m_1 + m_2)x$ और L का प्रतिच्छेदन बिंदु किस रेखा पर है :

**[JEE(Main)April,2023]**

(A) $6x + y = 10$ (B) $y - 2x = 5$ (C) $y - x = 5$ (D) $6x - y = 15$

**Ans.** C

**Sol.**




$$m_1 = \frac{3}{(10/3)} = \frac{9}{10}$$

$$m_2 = \frac{6}{(5/3)} = \frac{18}{5}$$

$$m_1 + m_2 = \frac{9+36}{10} = \frac{45}{10} = \frac{9}{2}$$

So, $y = (m_1 + m_2)x$ is $y = \frac{9}{2}x$

Now solving $y = \frac{9}{2}x$ and $9x + 5y = 45$

$$\Rightarrow 2y + 5y = 45 \Rightarrow y = \frac{45}{7}, x = \frac{10}{7}$$

$$y - x = \frac{35}{7} = 5$$

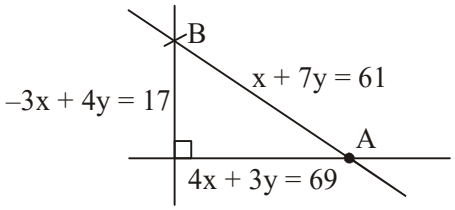
10. Let $C(\alpha, \beta)$ be the circumcenter of the triangle formed by the lines $4x + 3y = 69$, $4y - 3x = 17$, and $x + 7y = 61$. Then $(\alpha - \beta)^2 + \alpha + \beta$ is equal to : **[JEE(Main)April,2023]**

माना रेखाओ $4x + 3y = 69$, $4y - 3x = 17$, तथा $x + 7y = 61$ द्वारा निर्मित त्रिभुज का परिकेन्द्र $C(\alpha, \beta)$ है। तो $(\alpha - \beta)^2 + \alpha + \beta$ बराबर है : **[JEE(Main)April,2023]**

(A) 18 (B) 17 (C) 16 (D) 15

**Ans.** B

**Sol.**




$4x + 28y = 244$

$4x + 3y = 69$

$25y = 175$

$y = 7, x = 12$

$A(12, 7)$

$-3x + 4y = 17$

$3x + 21y = 183$

$25y = 200$

$y = 8, x = 5$

$B(5, 8)$

$\therefore$ Circumcenter

$$\alpha = \frac{17}{2} \beta = \frac{15}{2}$$

$\left(\frac{17}{2}, \frac{15}{2}\right)$

$(\alpha-\beta)^2+\alpha+\beta$

$1 + 16 = 17$

11. Let R be a rectangle given by the lines x = 0, x = 2, y = 0 and y = 5. Let A(α, 0) and B(0, β), α ∈ [0, 2] and β ∈ [0, 5], be such that the line segment AB divides the area of the rectangle R in the ratio 4 : 1. Then, the mid-point of AB lies on a : **[JEE(Main)April,2023]**

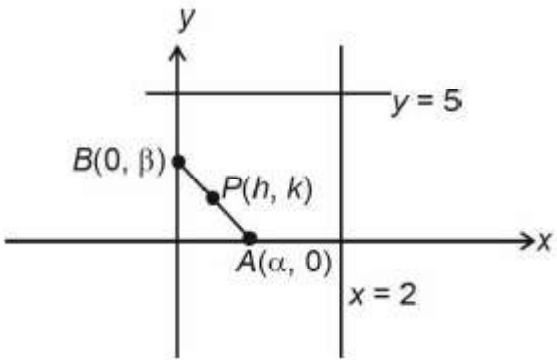
(A) circle (B) straight line (C) hyberbola (D) parabola

माना रेखाओं x = 0, x = 2, y = 0 तथा y = 5 द्वारा निर्मित आयत R है। माना A(α, 0) तथा B(0, β), α ∈ [0, 2], β ∈ [0, 5], इस प्रकार हैं कि रेखाखंड AB आयत R क्षेत्रफल को 4 : 1 के अनुपात में बाँटती है। तो AB का मध्य बिंदु एक **[JEE(Main)April,2023]**

(A) वृत्त (B) सरल रेखा (C) अतिपरवलय (D) परवलय

**Ans.** C

**Sol.**




$$\frac{10-\frac{1}{2}\alpha\beta}{\frac{1}{2}\alpha\beta}=\frac{4}{1} \Rightarrow 20-\alpha\beta=4\alpha\beta$$

$\Rightarrow \alpha\beta = 4$

Let $h=\frac{\alpha}{2}, \beta=\frac{k}{2}$

$\therefore \quad 4hk = 4$

$\therefore xy = 1$

12. If the point $\left(\alpha, \frac{7\sqrt{3}}{3}\right)$ lies on the curve traced by the mid-points of the line segments of the lines

$x\cos\theta + y\sin\theta = 7, \theta \in \left(0, \frac{\pi}{2}\right)$ between the co-ordinates axes, then $\alpha$ is equal to :

**[JEE(Main)April,2023]**

यदि रेखाओं $x\cos\theta + y\sin\theta = 7, \theta \in \left(0, \frac{\pi}{2}\right)$ के निर्देशांक अक्षों के बीच रेखाखंडों के मध्य बिंदुओं द्वारा बने वक्र पर एक बिंदु $\left(\alpha, \frac{7\sqrt{3}}{3}\right)$ है, तो $\alpha$ बराबर है : **[JEE(Main)April,2023]**

(A) $7\sqrt{3}$ (B) $-7\sqrt{3}$ (C) $-7$ (D) 7

**Ans.** D

**Sol.** $x\cos\theta + y\sin\theta = 7; \quad \theta \in \left(0, \frac{\pi}{2}\right)$

Mid point of line segment between coordinate axes

$$(h,k) \equiv \left(\frac{7}{2\cos\theta}, \frac{7}{2\sin\theta}\right)$$

$$\Rightarrow \text{Locus of (h, k) is } \frac{49}{y^2} + \frac{49}{x^2} = 4 \Rightarrow \frac{1}{x^2} + \frac{1}{y^2} = \frac{4}{49}$$

Point $\left(\alpha, \frac{7\sqrt{3}}{3}\right)$ lies on it

$$\Rightarrow \frac{1}{\alpha^2} + \frac{3}{49} = \frac{4}{49}$$

$$\Rightarrow \frac{1}{\alpha^2} = \frac{1}{49}$$

$$\Rightarrow \alpha = 7$$

13. Let $(\alpha, \beta)$ be the centroid of the triangle formed by the lines $15x - y = 82$, $6x - 5y = -4$ and $9x + 4y = 17$. Then $\alpha + 2\beta$ and $2\alpha - \beta$ are the roots of the equation : **[JEE(Main)April,2023]**

माना रेखाओं $15x - y = 82$, $6x - 5y = -4$ तथा $9x + 4y = 17$ द्वारा बने त्रिभुज का केन्द्रक $(\alpha, \beta)$ है। तो $\alpha + 2\beta$ तथा $2\alpha - \beta$ किस समीकरण के मूल हैं : **[JEE(Main)April,2023]**

(A) $x^2 - 7x + 12 = 0$ (B) $x^2 - 13x + 42 = 0$ (C) $x^2 - 10x + 25 = 0$ (D) $x^2 - 14x + 48 = 0$

**Ans.** B

**Sol.** $15x - y = 82$

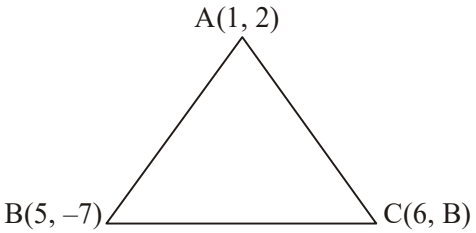
$6x - 5y = -4$

$9x + 4y = 17$

$(\alpha,\beta) \equiv \left(\frac{1+5+6}{3}, \frac{2-7+8}{3}\right) \equiv (4,1)$

$\alpha + 2\beta = 6$ and $2\alpha - \beta = 7$

Equation $x^2 - 13x + 42 = 0$



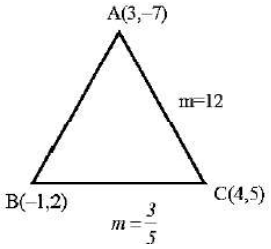

14. If $(\alpha, \beta)$ is the orthocenter of the triangle ABC with vertices A(3, –7), B(–1, 2) and C(4, 5), then $9\alpha - 6\beta + 60$ is equal to : **[JEE(Main)April,2023]**

यदि त्रिभुज ABC का लम्ब केन्द्र $(\alpha, \beta)$ है जहाँ त्रिभुज ABC के शीर्ष क्रमशः A(3, –7), B(–1, 2) तथा C(4, 5) हो, तो $9\alpha - 6\beta + 60$ का मान बराबर होगा : **[JEE(Main)April,2023]**

(A) 30 (B) 35 (C) 40 (D) 25

**Ans.** D

**Sol.**




Altitude of BC : $y + 7 = \frac{-5}{3}(x - 3)$

$3y + 21 = -5x + 15$

$5x + 3y + 6 = 0$

Altitude of AC : $y - 2 = \frac{-1}{12}(x + 1)$

$12y - 24 = -x - 1$

$x + 12y = 23$

$\alpha = \frac{-47}{19}, \beta = \frac{121}{57}$

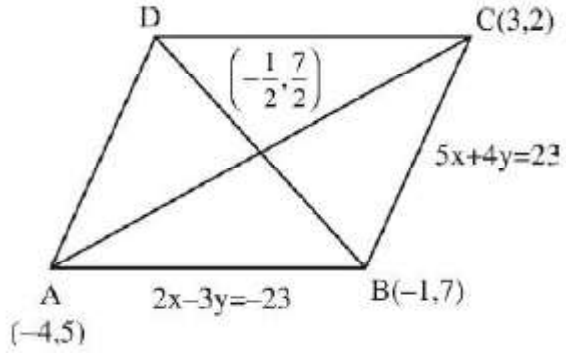
$9\alpha - 6\beta + 60 = 25$

15. Let the equations of two adjacent sides of a parallelogram ABCD be $2x - 3y = -23$ and $5x + 4y = 23$. If the equation of its one diagonal AC is $3x + 7y = 23$ and the distance of A from the other diagonal is d, then $50d^2$ is equal to _____. **[JEE(Main)April,2023]**

माना एक समांतर चतुर्भुज की दो संलग्न भुजाओं के समीकरण $2x - 3y = -23$ तथा $5x + 4y = 23$ हैं। यदि इसके एक विकर्ण AC का समीकरण $3x + 7y = 23$ है तथा A की दूसरे विकर्ण से दूरी d है, तो $50d^2$ बराबर है : **[JEE(Main)April,2023]**

**Ans.** 529

**Sol.**




A & C point will be (–4, 5) & (3, 2)

mid point of AC will be $\left(-\frac{1}{2},\frac{7}{2}\right)$

equation of diagonal BD is

$$y-\frac{7}{2}=\frac{\frac{7}{2}}{-\frac{1}{2}}\left(x+\frac{1}{2}\right)$$

$\Rightarrow 7x+y=0$

Distance of A from diagonal BD

$$=d=\frac{23}{\sqrt{50}}$$

$\Rightarrow 50d^2=(23)^2$

$50d^2 = 529$

16. If the line $l_1 : 3y - 2x = 3$ is the angular bisector of the lines $l_2 : x - y + 1 = 0$ and $l_3 : \alpha x + \beta y + 17 = 0$, then $\alpha^2 + \beta^2 - \alpha - \beta$ is equal to _____. **[JEE(Main)April,2023]**

यदि रेखा $l_1 : 3y - 2x = 3$ रेखाओं $l_2 : x - y + 1 = 0$ तथा $l_3 : \alpha x + \beta y + 17 = 0$ की कोण समद्विभाजक है, तो $\alpha^2 + \beta^2 - \alpha - \beta$ बराबर है _____ **[JEE(Main)April,2023]**

**Ans.** 348

**Sol.** Point of intersection of $\ell_1 : 3y - 2x = 3$

$\ell_2 : x - y + 1 = 0$ is $P \equiv (0,1)$

Which lies on $\ell_3 : \alpha x + \beta y + 17 = 0$

$\Rightarrow \beta = -17$

Consider a random point $Q \equiv (-1,0)$

on $\ell_2 : x - y + 1 = 0$, image of Q about

$\ell_2 : x - y + 1 = 0$, is $Q' \equiv \left(\frac{-17}{13}, \frac{6}{13}\right)$ which is calculated by formulae

$$\frac{x-(-1)}{2} = \frac{y-0}{-3} = 2\left(\frac{-2+3}{13}\right)$$

Now, Q' lies in $\ell_3 : \alpha x + \beta y + 17 = 0$

$\Rightarrow \alpha = 7$

Now, $\alpha^2 + \beta^2 - \alpha - \beta = 348$

17. Consider the triangle with vertices A(2, 1), B(0, 0) and C(t, 4), $t \in [0,4]$. If the maximum and the minimum perimeters of such triangles are obtained at $t = \alpha$ and $t = \beta$ respectively, then $6\alpha + 21\beta$ is equal to _____.
**[JEE(Main)April,2023]**

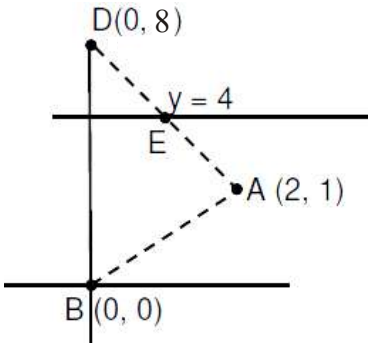
एक त्रिभुज इस प्रकार है जिसके शीर्ष A(2, 1), B(0, 0) तथा C(t, 4), $t \in [0,4]$ है। इस प्रकार के त्रिभुजों का अधिकतम तथा न्यूनतम परिमाप क्रमशः $t = \alpha$ तथा $t = \beta$ पर प्राप्त होता है तो $6\alpha + 21\beta$ का मान होगा _____ **[JEE(Main)April,2023]**

**Ans.** 48

**Sol.** A (2, 1), B(0, 0), C (t, 4), $t \in [0, 4]$

Image of B in $y = 4$

D(0, 8)



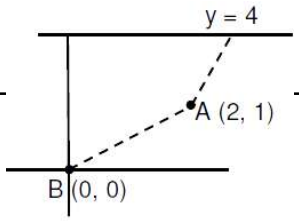

Equation of AD :

$$y - 1 = \frac{7}{-2}(x - 2)$$

$7x + 2y = 16$

$$E\left(\frac{8}{7}, 4\right) \Rightarrow \beta = \frac{8}{7}$$

For maximum perimeter



Maximum perimeter at C (4, 4)

$\therefore t \in [0,4]$

Now, $6\alpha + 21\beta = 6\times 4 + 21\left(\frac{8}{7}\right) = 48$

18. The portion of the line $4x + 5y = 20$ in the first quadrant is trisected by the lines $L_1$ and $L_2$ passing through the origin. The tangent of an angle between the lines $L_1$ and $L_2$ is : **[JEE(Main)Jan.2024]**

रेखा $4x + 5y = 20$ के प्रथम चतुर्थांश में भाग को मूल बिन्दु से होकर जाने वाली रेखाएँ $L_1$ तथा $L_2$ समत्रिभाजित करती हैं। $L_1$ तथा $L_2$ के बीच एक कोण की स्पर्शज्या है : **[JEE(Main)Jan.2024]**

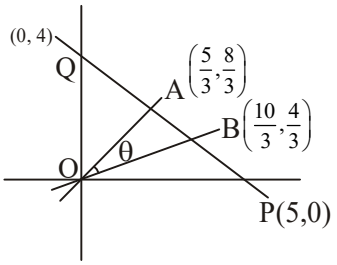
(A) $\frac{8}{5}$ (B) $\frac{2}{5}$ (C) $\frac{30}{41}$ (D) $\frac{25}{41}$

**Ans.** C

**Sol.** $A\left(\frac{5}{3},\frac{8}{3}\right)$, $B\left(\frac{10}{3},\frac{4}{3}\right)$

$m_{OA} = m_2 = \frac{8}{5}$

$m_{0B} = m_1 = \frac{2}{5}$




$$\tan\theta = \left|\frac{m_2 - m_1}{1 + m_1 m_2}\right| = \left|\frac{\frac{8}{5} - \frac{2}{5}}{1 + \frac{8}{5}\times\frac{2}{5}}\right|$$

$$= \frac{\frac{6}{5}}{1 + \frac{16}{25}} = \frac{6\times 5}{25 + 16} = \frac{30}{41}$$

19. Let R be the interior region between the lines $3x - y + 1 = 0$ and $x + 2y - 5 = 0$ containing the origin. The set of all values of a, for which the points $\left(a^2, a+1\right)$ lie in R, is : **[JEE(Main)Jan.2024]**

माना रेखाओं $3x - y + 1 = 0$ तथा $x + 2y - 5 = 0$ के अंदर का क्षेत्र, जिसमें मूलबिन्दु है, R है। a के सभी मानों, जिनके लिए $\left(a^2, a+1\right)$ क्षेत्र R में है, का समुच्चय है : **[JEE(Main)Jan.2024]**

(A) $(-3,-1)\cup\left(-\frac{1}{3},1\right)$ (B) $(-3,-1)\cup\left(\frac{1}{3},1\right)$

(C) $(-3,0)\cup\left(\frac{1}{3},1\right)$ (D) $(-3,0)\cup\left(\frac{2}{3},1\right)$

**Ans.** C

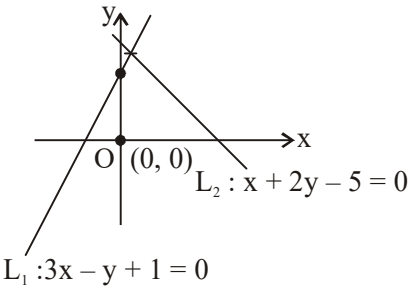
**Sol.** $P(a^2, a + 1)$

$L_1 = 3x - y + 1 = 0$

Origin and P lies same side w.r.t. $L_1$

$\Rightarrow L_1(0) \cdot L_1(P) > 0$

$\therefore 3(a^2) - (a + 1) + 1 > 0$




$\Rightarrow 3a^2 - a > 0$

$a \in (-\infty, 0) \cup \left(\frac{1}{3}, \infty\right)$ _____(1)

Let $L_2 : x + 2y - 5 = 0$

Originand P lies same side w.r.t. $L_2$

$\Rightarrow L_2(0) \cdot L_2(P) > 0$

$\Rightarrow a^2 + 2(a + 1) - 5 < 0$

$\Rightarrow a^2 + 2a - 3 < 0$

$\Rightarrow (a + 3)(a - 1) < 0$

$\therefore a \in (-3, 1)$ _____(2)

Intersection of (1) and (2)

$a \in (-3, 0) \cup \left(\frac{1}{3}, 1\right)$

20. Let $\left(5, \frac{a}{4}\right)$ be the circumcenter of a triangle with vertices A(a, –2), B(a, 6) and $C\left(\frac{a}{4}, -2\right)$. Let $\alpha$ denote the circumradius, $\beta$ denote the area and $\gamma$ denote the perimeter of the triangle. Then $\alpha + \beta + \gamma$ is :

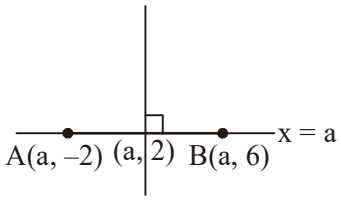
**[JEE(Main)Jan.2024]**

माना A(a, –2), B(a, 6) तथा $C\left(\frac{a}{4}, -2\right)$ के एक त्रिभुज का परिकेन्द्र $\left(5, \frac{a}{4}\right)$ है। यदि इस त्रिभुज के परिवृत्त की त्रिज्या $\alpha$, त्रिभुज का क्षेत्रफल $\beta$ तथा त्रिभुज का परिमाप $\gamma$ है, तो $\alpha + \beta + \gamma$ is बराबर है : **[JEE(Main)Jan.2024]**

(A) 53 (B) 60 (C) 62 (D) 30

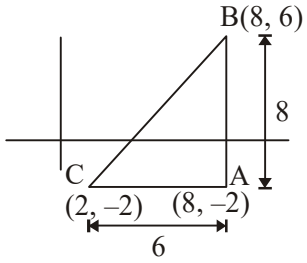
**Ans.** A

**Sol.**




Circumcentre lie on perpendicular bisector of AB $\Rightarrow y = 2$

$\Rightarrow \frac{a}{4} = 2$ $\Rightarrow a = 8$




$\Rightarrow$ Perimeter = 6 + 8 + 10 = 24

Area = $\frac{6\times 8}{2} = 24$

Circumradius = half of hypotenuse

$= \frac{10}{2} = 5$

$\alpha + \beta + \gamma = 24 + 24 + 5 = 53$

21. In a $\Delta ABC$, suppose $y = x$ is the equation of the bisector of the angle B and the equation of the side AC is $2x - y = 2$. If $2AB = BC$ and the points A and B are respectively (4, 6) and $(\alpha, \beta)$, then $\alpha + 2\beta$ is equal to :

**[JEE(Main)Jan.2024]**

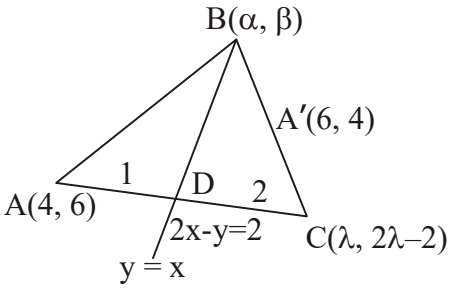
एक $\Delta ABC$ में माना कोण B के समद्विभाजक का समीकरण $y = x$ है तथा भुजा AC का समीकरण $2x - y = 2$ है। यदि (4, 6) और $(\alpha, \beta)$ क्रमशः बिन्दु A और B तथा $2AB = BC$ है, तो $\alpha + 2\beta$ बराबर है :

**[JEE(Main)Jan.2024]**

(A) 48 (B) 39 (C) 45 (D) 42

**Ans.** D

**Sol.**




$$\frac{AB}{BC}=\frac{AD}{DC}=\frac{1}{2}$$

$$D\left(\frac{\lambda+8}{3},\frac{2\lambda-2+12}{3}\right) \text{ lies on } y=x$$

$$\frac{\lambda+8}{3}=\frac{2\lambda+10}{3}$$

$$\Rightarrow \lambda=-2 \Rightarrow c(-2,-6)$$

Image of A w.r.t. y = x lies on BC

m(BC) = m(A'C)

$$\Rightarrow \frac{\alpha+6}{\alpha+2}=\frac{4+6}{6+2} \Rightarrow \alpha=14=\beta$$

$\alpha + 2\beta = 42$

22. The distance of the point (2,3) from the line $2x - 3y + 28 = 0$, measured parallel to the line $\sqrt{3}x - y + 1 = 0$, is equal to : **[JEE(Main)Jan.2024]**

बिन्दु (2, 3) की रेखा $2x - 3y + 28 = 0$ से रेखा $\sqrt{3}x - y + 1 = 0$ के समांन्तर मापी दूरी बराबर है : **[JEE(Main)Jan.2024]**

(A) $4+6\sqrt{3}$ (B) $4\sqrt{2}$ (C) $6\sqrt{3}$ (D) $3+4\sqrt{2}$

**Ans.** A

**Sol.** Given line $y = \sqrt{3}x + 1$

$$\text{Inclination} = \frac{\pi}{3}$$

Any point on line though (2,3) and parallel to given line.

$$P\left(2+r\cos\frac{\pi}{3},3+r\sin\frac{\pi}{3}\right)\equiv\left(2+\frac{r}{2},3+\frac{r\sqrt{3}}{2}\right)$$

If P lieson $2x - 3y + 28 = 0$

$$2\left(2+\frac{r}{2}\right)-3\left(3+\frac{r\sqrt{3}}{2}\right)+28=0$$

$\Rightarrow |r| = 4+6\sqrt{3}$ = Required distance

23. Let A be the point of intersection of the lines $3x+2y=14$, $5x-y=6$ and B be the point of intersection of the lines $4x+3y=8$, $6x+y=5$. The distance of the point P(5,–2) from the line AB is : **[JEE(Main)Jan.2024]**

माना रेखाओं $3x+2y=14$, $5x-y=6$ का प्रतिच्छेदन बिन्दु A है तथा रेखाओं $4x+3y=8$, $6x+y=5$ का प्रतिच्छेदन बिन्दु B है। तो रेखा AB से बिन्दु P(5,–2) की दूरी है : **[JEE(Main)Jan.2024]**

(A) $\frac{5}{2}$ (B) 8 (C) $\frac{13}{2}$ (D) 6

**Ans.** D

**Sol.** Solving lines $L_1$ $(3x+2y=14)$ and $L_2$ $(5x-y=6)$

to get A(2, 4) and solving lines $L_3$ $(4x+3y=8)$

and $L_4$ $(6x+y=5)$ to get $B\left(\frac{1}{2},2\right)$

Finding Eqn. of AB : $4x-3y+4=0$

Calculate distance PM $\Rightarrow \left|\frac{4(5)-3(-2)+4}{5}\right| = 6$

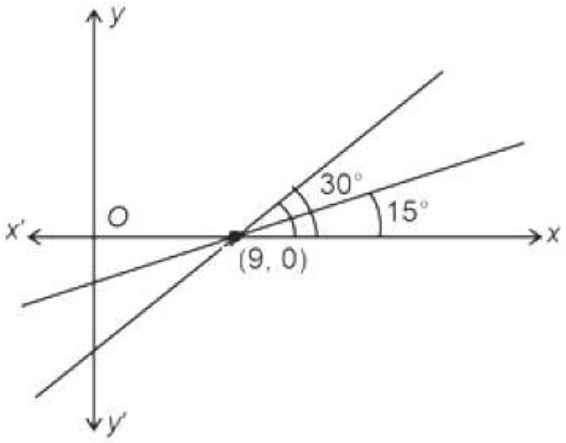
24. A line passing through the point A (9, 0) makes an angle of 30º with the positive direction of x-axis. If this line is rotated about A through an angle of 15º in the clockwise direction, then its equation in the new position is : **[JEE(Main)Jan.2024]**

बिंदु A (9, 0) से होकर जाने वाली एक रेखा x-अक्ष की धनात्मक दिशा से 30º का कोण बानाती है। यदि इस रेखा को घड़ी की दिशा में 15º के कोण तक A के सापेक्ष घुमाया जाता है, तो नयी स्थिति में इसका समीकरण है : **[JEE(Main)Jan.2024]**

(A) $\frac{x}{\sqrt{3}-2}+y=9$ (B) $\frac{x}{\sqrt{3}+2}+y=9$ (C) $\frac{y}{\sqrt{3}-2}+x=9$ (D) $\frac{y}{\sqrt{3}+2}+x=9$

**Ans.** C

**Sol.**




Equation : $y-0=\tan 15^\circ (x-9) \Rightarrow y=\left(2-\sqrt{3}\right)(x-9)$

25. If $x^2-y^2+2hxy+2gx+2fy+c=0$ is the locus of a point, which moves such that it is always equidistant from the lines $x+2y+7=0$ and $2x-y+8=0$, then the value of $g+c+h-f$ equals : **[JEE(Main)Jan.2024]**

यदि एक बिन्दु, जो इस प्रकार चलता है कि इसकी रेखाओं $x+2y+7=0$ तथा $2x-y+8=0$ से दूरी बराबर रहती है, का बिन्दुपथ $x^2-y^2+2hxy+2gx+2fy+c=0$ है, तो $g+c+h-f$ का मान बराबर है : **[JEE(Main)Jan.2024]**

(A) 6 ( 2) 8 (C) 29 (D) 14

**Ans.** D

**Sol.** Locus of point P(x, y) whose distance from

Gives

X + 2y + 7 = 0 & 2x – y + 8 = 0 are equal is $\frac{x+2y+7}{\sqrt{5}}=\pm\frac{2x-y+8}{\sqrt{5}}$

$(x+2y+7)^2-(2x-y+8)^2=0$

Combined equation of lines

$(x-3y+1)(3x+y+15)=0$

$3x^2-3y^2-8xy+18x-44y+15=0$

$x^2-y^2-\frac{8}{3}xy+6x-\frac{44}{3}y+5=0$

$x^2-y^2+2h\,xy+2gx+2fy+c=0$

$h=\frac{4}{3},\ g=3,\ f=-\frac{22}{3},\ c=5$

$g+c+h-f=3+5-\frac{4}{3}+\frac{22}{3}=8+6=14$

26. Let $\alpha, \beta, \gamma, \delta \in Z$ and let $A(\alpha, \beta)$, $B(1,0)$, $C(\gamma, \delta)$ and $D(1,2)$ be the vertices of a parallelogram ABCD. If $AB = \sqrt{10}$ and the points A and C lie on the line $3y = 2x + 1$, then $2(\alpha + \beta + \gamma + \delta)$ is equal to :

**[JEE(Main)Jan.2024]**

माना $\alpha, \beta, \gamma, \delta \in Z$ हैं तथा माना एक समांतर चतुर्भज ABCD के शीर्ष $A(\alpha, \beta)$, $B(1,0)$, $C(\gamma, \delta)$ तथा $D(1,2)$ हैं। यदि $AB = \sqrt{10}$ है तथा बिन्दु A और C रेखा $3y = 2x + 1$ पर हैं, तो $2(\alpha + \beta + \gamma + \delta)$ बराबर है :

**[JEE(Main)Jan.2024]**

(A) 12 (B) 10 (C) 5 (D) 8

**Ans.** D

**Sol.** ABCD is a parallelogram so mid-point of AC = mid-point of BD

$$\Rightarrow \left(\frac{\alpha+\gamma}{2}, \frac{\beta+\delta}{2}\right) = \left(\frac{1+1}{2}, \frac{0+2}{2}\right)$$

$$\Rightarrow \left(\frac{\alpha+\gamma}{2}, \frac{\beta+\delta}{2}\right) = (1,1)$$

$$\Rightarrow \alpha+\gamma = 2 \text{ and } \beta+\delta = 2$$

So, $2(\alpha+\beta+\gamma+\delta) = 2(2+2) = 8$

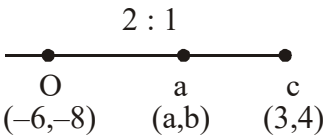
27. Let A(a, b), B(3, 4) and C(–6, –8) respectively denote the centroid, circumcentre and orthocentre of a triangle. Then, the distance of the point P(2a + 3, 7b + 5) from the line $2x + 3y - 4 = 0$ measured parallel to the line $x - 2y - 1 = 0$ is : **[JEE(Main)Jan.2024]**

माना A(a, b), B(3, 4) तथा C(–6, –8) एक त्रिभुज के केन्द्रक, परिकेन्द्र तथा लंबकेन्द्र है। तो बिन्दु P(2a + 3, 7b + 5) की रेखा $2x + 3y - 4 = 0$ से, रेखा $x - 2y - 1 = 0$ के समांतर नापी गई दूरी है : **[JEE(Main)Jan.2024]**

(A) $\frac{17\sqrt{5}}{6}$ (B) $\frac{17\sqrt{5}}{7}$ (C) $\frac{\sqrt{5}}{17}$ (D) $\frac{15\sqrt{5}}{7}$

**Ans.** B

**Sol.** Centroid divides orthocentre and circumcentre in 2 : 1.

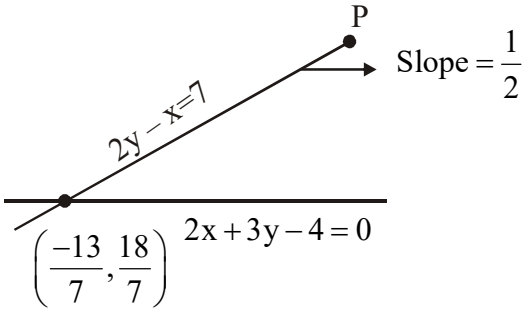



$$a \equiv \frac{2(3)-6}{3} = 0$$

$$b \equiv \frac{2(4)-8}{3} = 0$$

$$\Rightarrow (a, b) \equiv (0, 0)$$

$\Rightarrow P \equiv (3,5)$




$\Rightarrow (y-5)=\frac{1}{2}(x-3)$

$\Rightarrow 2y-10=x-3$

$\Rightarrow 2y-x=7$

$\Rightarrow \sqrt{\left(3+\frac{13}{7}\right)^2+\left(5-\frac{18}{7}\right)^2}=\frac{\sqrt{34^2+17^2}}{7}$

$=\frac{17\sqrt{5}}{7}$

28. If the sum of squares of all real values of $\alpha$, for which the lines $2x - y + 3 = 0$, $6x + 3y + 1 = 0$ and $\alpha x + 2y - 2 = 0$ do not form a triangle is p, then the greatest integer less than or equal to p is _____.

**[JEE(Main)Jan.2024]**

माना $\alpha$ के सभी वास्तविक मानों, जिनके लिए रेखाएँ $2x - y + 3 = 0$, $6x + 3y + 1 = 0$ तथा $\alpha x + 2y - 2 = 0$ एक त्रिभुज नहीं बनाती हैं, के वर्गों का योग p है, तो महत्तम पूर्णांक $\leq$ p है _____ **[JEE(Main)Jan.2024]**

**Ans.** 32

**Sol.** $2x - y + 3 = 0$

$6x + 3y + 1 = 0$

$\alpha x + 2y - 2 = 0$

will not form a $\Delta$ if $\alpha x + 2y - 2 = 0$ is concurrent with $2x - y + 3 = 0$ and $6x + 3y + 1 = 0$ or parallel to either of them so

Case-1: Concurrent lines

$$\begin{vmatrix} 2 & -1 & 3 \\ 6 & 3 & 1 \\ \alpha & 2 & -2 \end{vmatrix} = 0 \Rightarrow \alpha = \frac{4}{5}$$

Case-2 : Parallel lines

$$-\frac{\alpha}{2}=\frac{-6}{3} \text{ or } -\frac{\alpha}{2}=2$$

$\Rightarrow \alpha = 4$ or $\Rightarrow \alpha = -4$

$$P = 16+16+\frac{16}{25}$$

$$[P]=\left[32+\frac{16}{25}\right]=32$$

29. Let for any three distinct consecutive terms a, b, c of an A.P., the lines $ax + by + c = 0$ be concurrent at the point P and Q $(\alpha, \beta)$ be a point such that the system of equations

$x + y + z = 6$,

$2x + 5y + \alpha z = \beta$ and

$x + 2y + 3z = 4$, has infinitely many solutions. Then $(PQ)^2$ is equal to _____ .

**[JEE(Main)Jan.2024]**

माना एक A.P. के किसी भी तीन भिन्न क्रमागत पदों a, b, c के लिए रेखाऐं $ax + by + c = 0$ एक बिन्दु P पर संगामी हैं तथा बिन्दु Q $(\alpha, \beta)$ के लिए समीकरण निकाय

$x + y + z = 6$,

$2x + 5y + \alpha z = \beta$ तथा

$x + 2y + 3z = 4$, के अनंत हल है। तो $(PQ)^2$ बराबर है _____ . **[JEE(Main)Jan.2024]**

**Ans.** 113

**Sol.** $b - a = c - b$

$a - 2b + c = 0$

$\Rightarrow$ $ax + by + c = 0$ are concurrent at P $(1, -2)$

$\therefore$ $x + y + z = 6$,

$2x + 5y + \alpha z = \beta$,

$x + 2y + 3z = 4$

$\Delta = 8 - \alpha$,

$\Delta_1 = -8\alpha - \beta + 70$,

$\Delta_2 = 2\alpha + 2\beta - 28$,

$\Delta_3 = 6 - \beta$ for infinitely many solutions

$\Delta = \Delta_1 = \Delta_2 = \Delta_3 = 0$

$\Rightarrow \alpha = 8, \beta = 6$

$\Rightarrow Q(8, 6), P(1, -2)$

$\Rightarrow (PQ)^2 = 113$

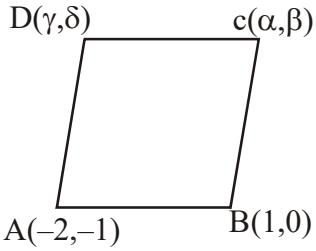
30. Let A(–2, –1), B(1, 0), C($\alpha$, $\beta$) and D($\gamma$, $\delta$) be the vertices of a parallelogram ABCD. If the point C lies on $2x - y = 5$ and the point D lies on $3x - 2y = 6$, then the value of $|\alpha + \beta + \gamma + \delta|$ is equal to _____.

**[JEE(Main)Jan.2024]**

माना एक समांतर चतुर्भुज ABCD के शीर्ष A(–2, –1), B(1, 0), C($\alpha$, $\beta$) तथा D($\gamma$, $\delta$) हैं। यदि बिन्दु C, रेखा $2x - y = 5$ पर है तथा बिन्दु D, रेखा $3x - 2y = 6$ पर है, तो $|\alpha + \beta + \gamma + \delta|$ का मान बराबर है _____ **[JEE(Main)Jan.2024]**

**Ans.** 32

**Sol.**




$2\alpha - \beta = 5$ .....(i)

$3\gamma - 2\delta = 6$ .....(ii)

Diagonal bisect each other

$\alpha - 2 = \gamma + 1 \Rightarrow \alpha = \gamma + 3$

$\beta - 1 = \delta \quad \Rightarrow \beta = \delta + 1$

Now form (i)

$2(\gamma + 3) - \delta - 1 = 5 \Rightarrow \delta = 2\gamma$

From (ii) $\gamma = -6, \quad \delta = -12, \quad \alpha = 3, \quad \beta = -11$

$|\alpha + \beta + \gamma + \delta| = 32$

31. Let ABC be an isosceles triangle in which A is at (–1, 0), $\angle A = \frac{2\pi}{3}$, AB = AC and B is on the positive x-axis. If $BC = 4\sqrt{3}$ and the line BC intersects the line $y = x + 3$ at ($\alpha$, $\beta$), then $\frac{\beta^4}{\alpha^2}$ is _____.

**[JEE(Main)Jan.2024]**

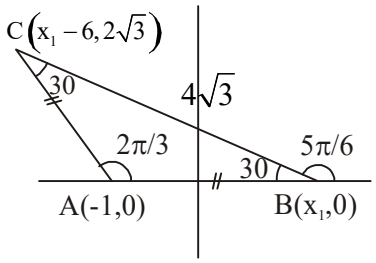
माना एक समद्विबाहु त्रिभुज ABC में A, बिंदु (–1, 0) हैं, $\angle A = \frac{2\pi}{3}$ हैं,, AB = AC है तथा B, धनात्मक x-अक्ष पर है। यदि

$BC = 4\sqrt{3}$ तथा रेखा BC, रेखा $y = x + 3$ को $(\alpha, \beta)$ पर काटती है, तो $\frac{\beta^4}{\alpha^2}$ बराबर है _____

**[JEE(Main)Jan.2024]**

**Ans.** 36

**Sol.**




Co-ordinates of 'C' are

$$\frac{x - x_1}{\cos\frac{5\pi}{6}} = \frac{y - 0}{\sin\frac{5\pi}{6}} = 4\sqrt{3}$$

$$x = 4\sqrt{3}\cos\frac{5\pi}{6} + x_1 \Rightarrow 4\sqrt{3} \times \left(-\frac{\sqrt{3}}{2}\right) + x_1 \Rightarrow (x_1 - 6)$$

$$y = 4\sqrt{3}\sin\frac{5\pi}{6} \Rightarrow 4\sqrt{3} \times \frac{1}{2} \Rightarrow 2\sqrt{3}$$

$$C\left(x_1 - 6, 2\sqrt{3}\right)$$

$$m_{AC} = -\sqrt{3}$$

$$\frac{0 - 2\sqrt{3}}{-1 - x_1 + 6} = -\sqrt{3} \Rightarrow x_1 = 3$$

$$BC : x + \sqrt{3}y = 3, y = x + 3$$

$$y - 3 + \sqrt{3}y = 3$$

$$y = \frac{6}{\sqrt{3}+1}, x = \frac{6}{\sqrt{3}+1} - 3 \Rightarrow \frac{6 - 3\sqrt{3} - 3}{\sqrt{3}+1}$$

$$x = \frac{3 - 3\sqrt{3}}{\sqrt{3}+1}$$

$$\frac{-6}{(1+\sqrt{3})^2}$$

So, $\frac{\beta^4}{\alpha^2} = 36$

32. The vertices of a triangle are A(–1,3), B(–2,2) and C(3, –1). A new triangle is formed by shifting the sides of the triangle by one unit inwards. Then the equation of the side of the new triangle nearest to origin is :

**[JEE(Main)April,2024]**

एक त्रिभुज के शीर्ष A(–1,3), B(–2,2) तथा C(3, –1) हैं। इस त्रिभुज की भुजाओं को त्रिभुज के अंदर की तरफ एक इकाई स्थानांतरित कर एक नया त्रिभुज बनाया गया है। तो नए त्रिभुज की मूलबिन्दु के सबसे निकट भुजा का समीकरण है :

**[JEE(Main)April,2024]**

(A) $x + y - \left(2 - \sqrt{2}\right) = 0$

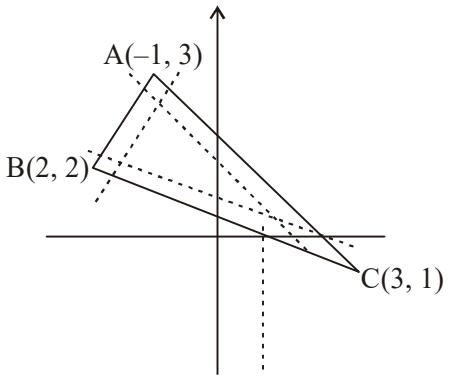
(B) $x - y - \left(2 + \sqrt{2}\right) = 0$

(C) $-x + y - \left(2 - \sqrt{2}\right) = 0$

(D) $x + y + \left(2 - \sqrt{2}\right) = 0$

**Ans.** A

**Sol.**




$AB : y - 2 = \frac{1}{1}(x + 2)$

$x - y - 4 = 0$ . $x - y + c = 0$

$\left|\frac{c+4}{\sqrt{2}}\right| = 1$

$C = \sqrt{2} - 4, -\sqrt{2} - 4$

$BC : \ y + 1 = \frac{3}{-5}(x - 3)$

$-5y - 5 = 3x - 9$

$3x + 5y = 4$

$\left|\frac{c+4}{\sqrt{34}}\right| = 1$

AC : $y-3=-1(x+1)$

$x+y-2$

$$\left|\frac{c+2}{\sqrt{2}}\right|=1$$

$C=\sqrt{2}-2,-\sqrt{2}-2$

Nearest : $x+y-2+\sqrt{2}=0$

33. Let two straight lines drawn from the origin O intersect the line $3x+4y=12$ at the points P and Q such that $\Delta OPQ$ is an isosceles triangle and $\angle POQ=90^{\circ}$. If $l=OP^2+PQ^2+QO^2$, then the greatest integer less than or equal to $l$ is : **[JEE(Main)April,2024]**

माना मूलबिन्दु से खींची गई दो रेखाएँ, रेखा $3x+4y=12$ को बिन्दुओं P तथा Q पर इस प्रकार काटती है कि $\Delta OPQ$ एक समद्विबाहु त्रिभुज है तथा $\angle POQ=90^{\circ}$ है। यदि $l=OP^2+PQ^2+QO^2$ है, तो $l$ से कम या $l$ के बराबर महत्तम पूर्णांक है :

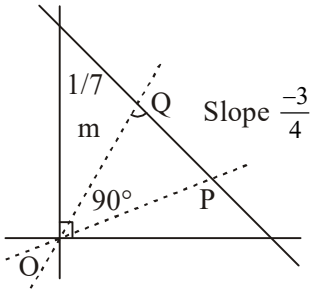
**[JEE(Main)April,2024]**

(A) 44 (B) 42 (C) 46 (D) 48

**Ans.** C

**Sol.**

$$\left|\frac{m+\frac{3}{4}}{1-\frac{3m}{4}}\right|=1$$




$$m+\frac{3}{4}=1-\frac{3}{4}m$$

$$m+\frac{3}{4}m=1-\frac{3}{4}$$

$7m=1$

$$\boxed{m=\frac{1}{7}}$$

$OQ = y=\frac{1}{7}x$ OP $y=7x$

$Q = 3x+4y=12$

$3x+\frac{4}{7}x=12$

$$x=\frac{12\times7}{25}, y=\frac{12\times7}{25\times7}=\frac{12}{25}, Q\left(\frac{84}{25},\frac{12}{25}\right)$$

$P \Rightarrow -25x=12$

$$x = \frac{-12}{25}, y = \frac{84}{25}$$

$$P = \left(\frac{-12}{25}, \frac{84}{25}\right)$$

$$OQ = \sqrt{\frac{84^2}{25^2} + \frac{12^2}{25^2}}$$

$$OP = \sqrt{\frac{84^2}{25^2} + \frac{12^2}{25^2}}$$

$$PQ = \sqrt{2}\sqrt{\frac{84^2}{25^2} + \frac{12^2}{25^2}}$$

$\ell = OQ^2 + OP^2 + PQ^2$

$$= \frac{84^2}{25^2} \times 2 + \frac{12^2}{25^2} \times 2 + 2\left(\frac{84^2}{25^2} + \frac{12^2}{25^2}\right)$$

$$= 4 \times \frac{84^2}{25^2} + 4 \times \frac{12^2}{25^2}$$

$$\ell = 4 \times \frac{12^2}{25^2}\left(7^2 + 1\right) = \frac{144 \times 8}{25} = 46.08$$

$[\ell] = 46$

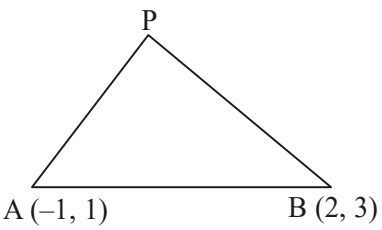
34. Let A (–1, 1) and B(2, 3) be two points and P be a variable point above the line AB such that the area of ΔPAB is 10. If the locus of P is ax + by = 15, then 5a + 2b is : **[JEE(Main)April,2024]**

माना दो बिन्दु A (–1, 1) तथा B(2, 3) हैं और रेखा AB के ऊपर एक चर बिन्दु P इस प्रकार है कि ΔPAB का क्षेत्रफल 10 है। यदि P का बिंदुपथ ax + by = 15 है, तो 5a + 2b बराबर है : **[JEE(Main)April,2024]**

(A) 6 (B) 4 (C) $-\frac{6}{5}$ (D) $-\frac{12}{5}$

**Ans.** D

**Sol.**




Locus of P,

$ax + by = 15$

Take point P as (h, k)

Area of $\Delta$ ABP

$= \frac{1}{2}|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)| = 10$

$= \frac{1}{2}|-1(3-k) + 2(k-1) + h(1-3)| = 10$

$= |-3 + k + 2k - 2 + h - 3h| = 20$

$= |3k - 2h - 5| = 20$

$= -(3k - 2h - 5) = 20 \Rightarrow 2h - 3k = 15$

$a = 2, b = -3$

$\therefore\ 5a + 2b = 4$

35. Let a variable line of slope $m > 0$ passing through the point $(4, -9)$ intersect the coordinate axes at the points A and B. The minimum value of the sum of the distances of A and B from the origin is : **[JEE(Main)April,2024]**

माना प्रवणता $m > 0$ तथा बिन्दु $(4, -9)$ से गुजरने वाली कोई चर रेखा निर्देशांक अक्षों को बिन्दु A और B पर प्रतिच्छेद करती है। मूल बिन्दु से A और B की दूरियों के योग का न्यूनतम मान है : **[JEE(Main)April,2024]**

(A) 30 (B) 25 (C) 15 (D) 10

**Ans.** B

**Sol.** Equation of line is

$y + 9 = m(x - 4)$

$\therefore A = \left(\frac{9 + 4m}{m}, 0\right)$

$B = (0, -9 - 4m)$

$\therefore OA + OB = \frac{9 + 4m}{m} + 9 + 4m$

$\because\ m > 0$

$= 13 + \frac{9}{m} + 4m$

$\because \frac{4m + \frac{9}{m}}{2} \geq \sqrt{36} \qquad \Rightarrow 4m + \frac{9}{m} \geq 12$

$\therefore\ OA + OB \geq 25$

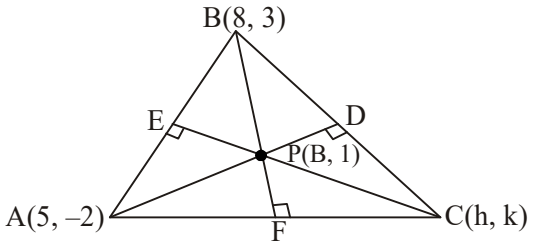
36. If P(6, 1) be the orthocentre of the triangle whose vertices are A(5, –2), B(8, 3) and C(h, k), then the point C lies on the circle : **[JEE(Main)April,2024]**

यदि P(6, 1) उस त्रिभुज का लंबकेन्द्र है, जिसके शीर्ष A(5, –2), B(8, 3) तथा C(h, k) हैं, तो बिन्दु C किस वृत्त पर स्थित है : **[JEE(Main)April,2024]**

(A) $x^2 + y^2 - 52 = 0$ (B) $x^2 + y^2 - 61 = 0$ (C) $x^2 + y^2 - 74 = 0$ (D) $x^2 + y^2 - 65 = 0$

**Ans.** D

**Sol.**




Solpe of BF $= (m_{BF}) = \frac{3-1}{8-6} = 1$

$m_{AC} = \frac{k+2}{h-5} - 1 (AC \perp BF)$

$\Rightarrow k + 2 = 5 - h$

$\Rightarrow k = 3 - h$ .....(i)

$m_{AD} = \frac{1+2}{6-5} = 3$

$m_{BC} = \frac{k-3}{h-8} = \frac{1}{3} (AD \perp BC)$

$\Rightarrow 3k - 9 = 8 - h$ .....(ii)

From (i) & (ii)

$h = -4, k = 7$

Now, $h^2 + k^2 = 16 + 49 = 65$

$h^2 + k^2 - 65 = 0$

Locus is $x^2 + y^2 - 65 = 0$

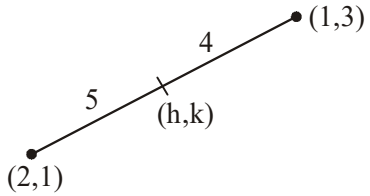
37. If the locus of the point, whose distances from the point (2, 1) and (1, 3) are in the ratio 5 : 4, is $ax^2 + by^2 + cxy + dx + ey + 170 = 0$, then the value of $a^2 + 2b + 3c + 4d + e$ is equal to : **[JEE(Main)April,2024]**

यदि उस बिन्दु, जिसकी बिन्दुओं (2, 1) और (1, 3) से दूरियों का अनुपात 5 : 4 है, का बिन्दुपथ $ax^2 + by^2 + cxy + dx + ey + 170 = 0$ है, तो $a^2 + 2b + 3c + 4d + e$ का मान बराबर है : **[JEE(Main)April,2024]**

(A) –27 (B) 37 (C) 5 (D) 437

**Ans.** B

**Sol.**




$$\therefore \frac{(h-2)^2+(k-1)^2}{(h-1)^2+(k-3)^2}=\frac{25}{16}$$

$$\frac{h^2+k^2-4h-2k+5}{h^2+k^2-2h-6k+10}=\frac{25}{16}$$

Replacing $h \to x$ and $k \to y$.

$16x^2 + 16y^2 - 64x - 32y + 80$

$= 25x^2 + 25y^2 - 50x - 150y + 250$

$9x^2 + 9y^2 + 14x - 118y + 170 = 0$

$\Rightarrow a = 9, b = 9, c = 0, d = 14, e = -118$

$a^2 + 2b + 3c + 4d + e$

$81 + 18 + 56 - 118$

$\Rightarrow 37$

38. The equations of two sides AB and AC of a triangle ABC are $4x + y = 14$ and $3x - 2y = 5$, respectively. The point $\left(2, -\frac{4}{3}\right)$ divides the third side BC internally in the ratio 2 : 1. The equation of the side BC is :

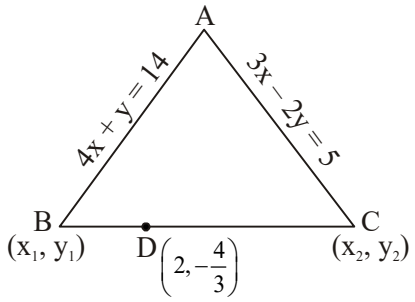
**[JEE(Main)April,2024]**

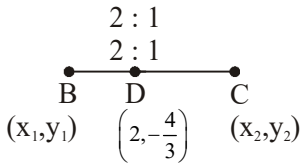
त्रिभुज ABC के दो भुजाओं AB तथा AC का समीकरण क्रमशः $4x + y = 14$ और $3x - 2y = 5$ है। यदि बिन्दु $\left(2, -\frac{4}{3}\right)$ तीसरी भुजा BC को 2 : 1 में अंतः विभाजित करता है, तो BC का समीकरण होगा : **[JEE(Main)April,2024]**

(A) $x + 3y + 2 = 0$ (B) $x + 6y + 6 = 0$ (C) $x - 6y - 10 = 0$ (D) $x - 3y - 6 = 0$

**Ans.** A

**Sol.**




$$2 = \frac{2x_2 + x_1}{3}, \frac{-4}{3} = \frac{2y_2 + y_1}{3}$$

$2x_2 + x_1 = 6,\ 2y_2 + y_1 = -4$

$x_1 = 6 - 2x_2$ ...(1)

$y_1 = -4 - 2y_2$ ...(2)

$4x_1 + y_1 = 14$ ...(3)

$3x_2 - 2y_2 = 15$ ...(4)

From here, $x_2 = 1,\ y_2 = -1,\ x_1 = 4,\ y_1 = -2$

B(4, –2) C(1, –1)

$$y + 2 = \frac{-1+2}{1-4}(x-4)$$

$-3y - 6 = x - 4$

$x + 3y + 2 = 0$

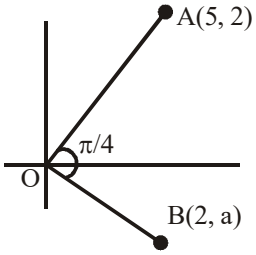
39. If the line segment joining the points (5,2) and (2,a) subtends an angle $\frac{\pi}{4}$ at the origin, then the absolute value of the product of all possible values of a is : **[JEE(Main)April,2024]**

यदि बिन्दुओं (5,2) तथा (2,a) को मिलाने वाला रेखाखंड, मूलबिन्दु पर $\frac{\pi}{4}$ का कोण बनाता है, तो a के सभी संभव मानों के गुणनफल का निरपेक्ष मान है : **[JEE(Main)April,2024]**

(A) 2 (B) 8 (C) 4 (D) 6

**Ans.** C

**Sol.**




$M_{OA} = \frac{2}{5}$

$M_{OB} = \frac{a}{2}$

$$= \tan\frac{\pi}{4} = \left|\frac{\frac{2}{5} - \frac{a}{2}}{1 + \frac{2}{5}\cdot\frac{a}{2}}\right|$$

$$\Rightarrow 1 = \left|\frac{4 - 5a}{10 + 2a}\right| \Rightarrow a = \frac{-6}{7}, \frac{14}{3}$$

Product $\Rightarrow -4$, Abs. value = 4

40. A ray of light coming from the point P(1, 2) gets reflected from the point Q on the x-axis and then passes through the point R(4, 3). If the point S(h, k) is such that PQRS is a parallelogram, then $hk^2$ is equal to :

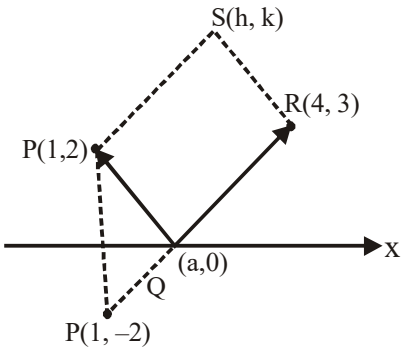
**[JEE(Main)April,2024]**

बिन्दु P(1, 2) से आने वाली रोशनी की एक किरण x-अक्ष पर बिन्दु Q से परावर्तित होने के पश्चात् बिन्दु R(4, 3) से गुजरती है। यदि बिन्दु S(h, k) इस प्रकार है कि PQRS एक समांतर चतुर्भुज है, तो $hk^2$ बराबर है : **[JEE(Main)April,2024]**

(A) 80 (B) 90 (C) 60 (D) 70

**Ans.** D

**Sol.**




$$P'R : y + 2 = \frac{5}{3}(x - 1)$$

For Point Q $\Rightarrow y = 0$

$$\frac{6}{5} = a - 1 \Rightarrow a = \frac{11}{5}$$

Now, PQRS is parallelogram

$\therefore \frac{h+a}{2}=\frac{4+1}{2} \Rightarrow h=5-\frac{11}{5}=\frac{14}{5}$

and $\frac{2+3}{2}=\frac{k}{2} \Rightarrow K=5$

Now $hk^2=25\times\frac{14}{5}=14\times5=70$

41. Two vertices of a triangle ABC are A(3,–1) and B(–2,3), and its orthocentre is P(1,1). If the coordinates of the point C are $(\alpha, \beta)$ and the centre of the circle circumscribing the triangle PAB is (h, k), then the value of $(\alpha+\beta)+2(h+k)$ equals : **[JEE(Main)April,2024]**

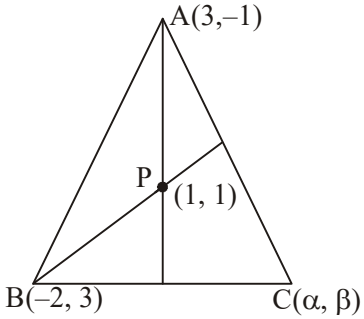
किसी त्रिभुज ABC के दो शीर्ष A(3,–1) और B(–2,3) हैं तथा उसका लंब केन्द्र P(1,1) है। यदि बिन्दु C के निर्देशांक $(\alpha, \beta)$ हैं तथा त्रिभुज PAB के परिवृत्त का केन्द्र (h, k) है, तब $(\alpha+\beta)+2(h+k)$ का मान बराबर है :

**[JEE(Main)April,2024]**

(A) 5 (B) 51 (C) 15 (D) 81

**Ans.** A

**Sol.**




Equation of PC : $y-1=\frac{5}{4}(x-1)\ \left(\because m_{AB}=-\frac{4}{5}\right)$

$4y-4=5x-5$

$5x-4y=1$ _____(1)

Equation of BC : $y-3=1(x+2)\ (\because m_{AP}=-1)$

$x-y=-5$ _____(2)

Solving equation (1) and (2), $x=21, y=26$

$\therefore \alpha=21, \beta=26 \Rightarrow (\alpha+\beta)=47$

Now, equation of $\perp$ bisector of AP : $y-0=1(x-2)$

$x-y=2$ _____(3)

Equation of $\perp$ bisector of AB : $y-1=\frac{5}{4}\left(x-\frac{1}{2}\right)$ _____(4)

Solving equation (3) and (4),

$x = -\frac{19}{2}, y = -\frac{23}{2}$

i.e $2h = -19, 2k = -23$

$\therefore\ 2(h + k) = -42$

So, $(\alpha + \beta) + 2(h + k) = 47 - 42 = 5$

42. Consider a triangle ABC having the vertices A(1,2), B($\alpha$, $\beta$) and C($\gamma$, $\delta$) and angles $\angle ABC = \frac{\pi}{6}$ and $\angle BAC = \frac{2\pi}{3}$. If the points B and C lie on the line $y = x + 4$, then $\alpha^2 + \gamma^2$ is equal to _____.

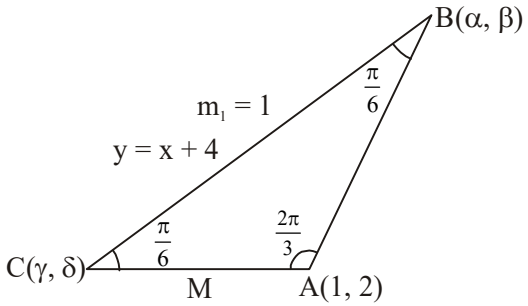
**[JEE(Main)April,2024]**

एक त्रिभुज ABC जिसके शीर्ष A(1,2), B($\alpha$, $\beta$) तथा C($\gamma$, $\delta$) और कोण $\angle ABC = \frac{\pi}{6}$ तथा $\angle BAC = \frac{2\pi}{3}$ हैं पर विचार कीजिए। यदि बिन्दु B तथा C रेखा $y = x + 4$ पर स्थित हैं, तो $\alpha^2 + \gamma^2$ बराबर है _____

**[JEE(Main)April,2024]**

**Ans.** 14

**Sol.**




Equation of line passing through P(1, 2) and making angle $\frac{\pi}{6}$ with $y = x + 4$ _____(1)

$y - 2 = m(x - 1)$

$y - 2 = \left(2 \pm \sqrt{3}\right)(x - 1)$ _____(2)

$\tan\frac{\pi}{6} = \left|\frac{m - m_1}{1 + mm_1}\right| \pm \frac{1}{\sqrt{3}} = \frac{m - 1}{1 + m}$

$m = 2 \pm \sqrt{3}$

Solve (1) and (2)

$x + 4 - 2 = \left(2 \pm \sqrt{3}\right)(x - 1)$

$x = \frac{4 + \sqrt{3}}{1 + \sqrt{3}} = \alpha$ and $x = \frac{4 - \sqrt{3}}{1 - \sqrt{3}} = \gamma$

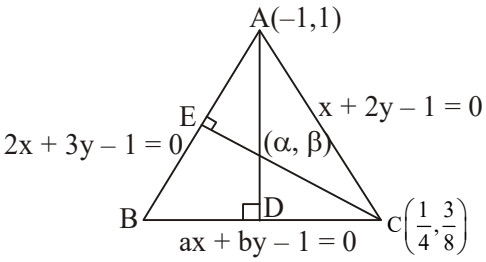
$\therefore\ \alpha^2 + \gamma^2 = 14$

43. If the orthocentre of the triangle formed by the lines $2x + 3y - 1 = 0$, $x + 2y - 1 = 0$ and $ax + by - 1 = 0$, is the centroid of another triangle, whose circumcentre and orthocentre respectively are (3, 4) and (–6, –8), then the value of $|a - b|$ is _____. **[JEE(Main)April,2024]**

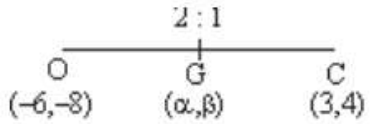
यदि रेखाओं $2x + 3y - 1 = 0$, $x + 2y - 1 = 0$ और $ax + by - 1 = 0$ द्वारा बनाया गया त्रिभुज, जिसका लंब केन्द्र किसी अन्य त्रिभुज के केन्द्रक के बराबर है एवं जिसका परिकेन्द्र और लंब केन्द्र क्रमशः (3, 4) तथा (–6, –8) है, तो $|a - b|$ का मान होगा _____ **[JEE(Main)April,2024]**

**Ans.** 16

**Sol.**




For another Δ




$$\alpha = \frac{2\times 3 - 6}{3} = 0 \qquad \beta = \frac{2\times 4 - 8}{3} = 0$$

$$(\alpha, \beta) = (0, 0)$$

$$x + 2y - 1 = 0$$
$$2x + 3y - 1 = 0$$
$$\underline{- \; - \quad +}$$
$$y - 1 = 0$$

$y = 1$

$x = -1$

A(–1, 1)

$$\text{Slope of AD} = \frac{0-1}{0+1} = -1$$

and

$M_{AD} \times M_{BC} = -1$

$(-1)\, M_{BC} = -1$

$M_{BC} = 1$

Now for point C

find equation of CE

$M_{CE} = \frac{2}{3}$

$y = \frac{2}{3}x$

and equation of AC

$$\begin{array}{r} x + 2y - 1 = 0 \\ -3x + 2y = 0 \\ \hline 4x - 1 = 0 \end{array}$$

$x = \frac{1}{4}$

$y = \frac{3}{2} \times \frac{1}{4} = \frac{3}{8}$

Eq. of BC

$C\left(\frac{1}{4}, \frac{3}{8}\right)$

$m_{BC} = 1$

$y - \frac{3}{8} = 1\left(x - \frac{1}{4}\right)$

$\frac{8y-3}{8} = \frac{4x-1}{4}$

$8y - 8x + 2 - 3 = 0$

$-8x + 8y - 1 = 0$

$ax + by - 1 = 0$

$a = -8$

$b = 8$

Value of $|a - b| = |-8 - 8| = 16$

44. Let a ray of light passing through the point (3,10) reflects on the line $2x + y = 6$ and the reflected ray passes through the point (7,2). If the equation of the incident ray is $ax + by + 1 = 0$, then $a^2 + b^2 + 3ab$ is equal to _____.

**[JEE(Main)April,2024]**

माना बिन्दु (3,10) से गुजरने वाली प्रकाश की कोई किरण रेखा $2x + y = 6$ पर परावर्तित होती है। तथा परावर्तित होती है तो परावर्तित किरण बिन्दु (7,2) से गुजरती है। यदि आपतित किरण समीकरण $ax + by + 1 = 0$ है, तो $a^2 + b^2 + 3ab$ बराबर है।

**[JEE(Main)April,2024]**

**Ans.** 1

**Sol.** $\frac{x-3}{2}=\frac{y-10}{1}=\frac{-2(6+10-6)}{4+1}=-4$

$x=-5,\ y=6$

$A'(-5,6),\ B(7,2)$

equation A'B $\quad y-6=\frac{2-6}{7+5}(x+5)$

$3y-18=-x-5$

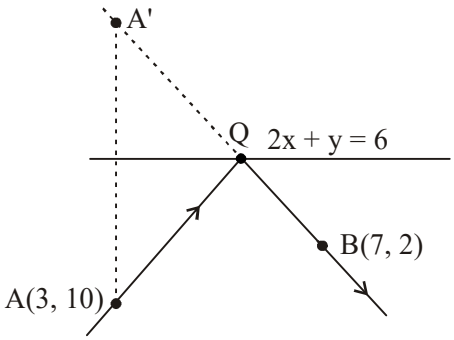
$x+3y=13$

$2x+6y=26$

$2x+y=6$ (given line)

$5y=20, \quad y=4,\ x=1$

$Q=(1,4)$




Required line $\quad y-10=\frac{4-10}{1-3}(x-3)$

$y-10=3x-9$

$3x-y+1=0$

$ax+by+1=0$

$a=3,\ b=-1$

$a^2+b^2+3ab=9+1-9=1$